

Analysis of Censorship Evasion Tools

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Abstract—Motivation: The growing prevalence of internet censorship by authoritarian regimes threatens global access to information and free expression. Despite tools like VPNs and Tor, evolving censorship techniques challenge their effectiveness. This study investigates the capabilities, limitations, and socio-political implications of these tools, aiming to support digital freedom and advance resilient evasion technologies.

Methods: This study employs technical analysis, tool testing, and case studies to evaluate censorship evasion tools like VPNs, Tor, and Psiphon. Metrics include performance, privacy, and obfuscation efficacy under diverse network conditions. Ethical and legal implications are also assessed to understand the tool’s effectiveness and impact in high-censorship regions.

Results: Censorship evasion tools exhibit varying effectiveness. NordVPN outperforms in speed and security, while Psiphon prioritizes accessibility but suffers performance issues. Tor excels in anonymity but has high latency. Testing reveals strengths in obfuscation and privacy, though limitations persist against advanced government countermeasures. Findings highlight trade-offs across tools and regions.

Conclusions: Censorship evasion tools are essential for preserving internet freedom in restrictive regimes. While effective, they face challenges like evolving countermeasures and performance trade-offs. Advancing adaptive technologies, improving accessibility, and addressing ethical concerns are critical for sustaining their impact on privacy, free expression, and socio-political empowerment in censored environments.

I. PROBLEM STATEMENT

As internet censorship becomes more widespread, particularly in authoritarian regimes, people face

increasing obstacles to accessing uncensored information and exercising free expression. Despite the existence of censorship evasion tools such as VPNs, Tor, and Psiphon, governments are continually developing and deploying countermeasures, including DNS filtering and geoblocking, to detect, block, or slow these tools. This situation raises questions about the effectiveness, resilience, and limitations of current censorship evasion tools.

This project addresses the challenge of understanding the landscape of censorship evasion technology and its limitations when confronted with censorship techniques employed by authoritarian regimes. In addition, we’ll examine ethical, legal, and security concerns surrounding the use of these tools, as they may be misused for illicit activities, which complicates their adoption and development. This study aims to explore these issues to better understand the effectiveness and limitations of censorship evasion tools and provide insight into the implications for freedom of speech, cybersecurity, and international governance.

II. BACKGROUND

Overview of Internet Censorship: Internet censorship refers to the deliberate restriction or suppression of information and communication on the Internet by governments, organizations, or other controlling bodies[1]. Implemented across various regions and contexts, censorship can take multiple forms, including blocking specific websites, filtering search results, restricting social media access, and monitoring online communications. While some countries impose censorship for reasons such as national security, prevention of hate speech, or copyright protection, in many cases, censorship is pri-

marily used by authoritarian regimes to control political dissent, prevent social mobilization, and restrict access to information that challenges official narratives[2].

Techniques for implementing internet censorship range from relatively straightforward measures, such as IP blocking and URL filtering, to more sophisticated approaches, like artificial intelligence-based traffic analysis [3]. Governments employ these techniques to identify and block specific types of traffic, often targeting services that enable users to bypass censorship, such as virtual private networks (VPNs), the Tor network, and encrypted messaging applications[4], [5]. Countries with extensive censorship infrastructures, such as China, Iran, and Russia, employ a combination of legal, technological, and infrastructural measures—such as China’s Great Firewall and Iran’s “Halal internet”—to maintain tight control over digital information flows[1].

The increasing trend of internet censorship poses significant challenges to individual freedoms, such as the right to freedom of speech, access to information, and privacy[2]. As governments adopt more advanced censorship techniques, users and organizations advocating for digital rights are forced to develop sophisticated methods to bypass restrictions and ensure open access to information [5]. The ongoing struggle between censorship and evasion demonstrates the critical role of technology in enabling access to information in the modern world[1].

Common Censorship Techniques: Governments employ various technical methods to restrict internet access and control the flow of information[1]. Key techniques include DPI, DNS filtering, IP blocking, and geoblocking. Each plays a specific role in monitoring and restricting online activity.

Deep Packet Inspection allows authorities to analyze the content of data packets being transmitted over the Internet. Using this technique, they can identify specific keywords, protocols, or applications[3]. By examining packets at a granular level, DPI can be used to detect and block traffic from censorship evasion tools like VPNs or Tor, which rely on encryption and unique data signatures[4]. For instance, DPI can recognize and filter out encrypted traffic by distinguishing it from typical

web traffic, preventing users from accessing restricted sites or bypassing censorship barriers[2].

DNS Filtering is another standard method, where access to specific websites is restricted by controlling the Domain Name System[1]. By manipulating DNS servers, governments can block users from resolving domain names associated with censored content or evasion tools, resulting in inaccessible sites or redirected requests[2]. This technique is frequently applied to block social media platforms, news websites, and circumvention tools that may facilitate uncensored communication[5].

IP Blocking involves restricting access to particular IP addresses associated with censored content or services[1]. By blocking these IP addresses, governments can prevent users within their jurisdiction from connecting to servers that host restricted websites or evasion tools[3]. This approach is practical but can result in over-blocking, where legitimate services that share IP addresses with restricted content are also rendered inaccessible[2].

Geoblocking targets users based on geographic location, limiting access to content or services from specific regions[2]. Often implemented with IP-based geolocation, geoblocking prevents users in certain countries from accessing foreign websites, apps, or news sources[1]. This technique is used for censorship and regulatory compliance, often coupled with IP blocking to restrict users’ access to uncensored information sources[5].

These techniques, especially when used together, enable governments to monitor, block, or slow specific types of internet traffic, making it difficult for users to bypass restrictions[1]. Advanced censorship infrastructures, such as China’s Great Firewall, deploy multiple techniques together to create a layered defense against censorship evasion tools, ensuring that even sophisticated methods of bypassing censorship are detected and obstructed [2]. The adaptive nature of these techniques means that as new tools emerge to circumvent censorship, government countermeasures evolve in response, maintaining control over internet access within restricted regions.

Introduction to Censorship Evasion Tools: Censorship evasion tools have evolved rapidly over the past

two decades in response to increasing internet restriction worldwide[2]. The most popular tools are VPNs, Tor, and Psiphon, each of which employs different mechanisms to evade censorship and enable users to access restricted content. VPNs, like NordVPN, encrypt internet traffic by creating a secure tunnel between the user and a server in another location[3]. This encrypted connection hides the user's IP address and masks their online activities, allowing them to bypass local censorship and access otherwise restricted websites and services[5].

The Tor network, another widely used tool, creates anonymity by routing traffic through a series of volunteer-operated nodes. This process, known as "onion routing," encrypts data multiple times, making it difficult for surveillance tools to trace the origin or destination of the traffic[4]. Psiphon combines VPN, SSH, and HTTP proxy technologies to evade censorship. It dynamically selects the best available method to bypass restrictions, adapting in real-time to changes in network conditions and censorship tactics[5].

These tools use encryption, obfuscation, and proxy networks to circumvent censorship[2]. Encryption ensures that data is secure from surveillance, while obfuscation techniques, like "stealth servers" in NordVPN, disguise encrypted traffic to make it appear like regular internet traffic, helping it evade detection by censorship system[3]. Proxy networks route traffic through external servers, hiding users' IP addresses and allowing them to appear as if they are browsing from another location[5]. These tools aim to preserve open internet access, enabling users in restricted regions to communicate freely, access unbiased information, and exercise their right to privacy online[1].

Current Challenges and Countermeasures: The relationship between censorship evasion tool developers and governments has been a continuous cycle of innovation and response, where each advancement in evasion technology prompts increasingly sophisticated countermeasures from authorities[1]. As VPNs, Tor, and similar tools evolve to bypass censorship, governments employ enhanced methods to detect, block, and slow these connections[2]. This cycle of adaptation and counter-adaptation creates a challenging environment for both

users and developers of censorship evasion tools[3].

Recent advances in censorship technology have greatly improved governments' ability to detect and block evasion tools[1]. One of the most notable developments is artificial intelligence-driven traffic analysis, which can identify patterns indicative of VPN or Tor traffic even when traditional DPI fails[3]. AI-driven systems can analyze large volumes of network traffic, recognizing and flagging characteristics that suggest encryption, obfuscation, or routing through proxy networks[2]. This technique enables authorities to target VPNs like NordVPN and other tools more effectively, even when they employ advanced obfuscation methods[5].

Additionally, advancements in DPI have made it possible to inspect encrypted traffic at a deeper level, identifying specific protocols and patterns associated with VPNs or Tor[4]. Modern DPI systems can differentiate VPN traffic from regular HTTPS traffic by analyzing packet sizes, timing, and other subtle variations, allowing governments to selectively block or throttle evasion tools without impacting general internet services[2]. These countermeasures highlight the adaptability and persistence of government censorship mechanisms, challenging developers to continuously refine their tools to stay ahead in the competition for open access[1].

Socio-Political Relevance: Censorship and evasion tools have profound socio-political implications, particularly in regions where internet restrictions curb freedom of speech, privacy, and digital rights[5]. Censorship limits individuals' access to information, controlling narratives and preventing exposure to diverse perspectives, which undermines informed public discourse[1]. Evasion tools such as VPNs, Tor, and Psiphon provide users with means to bypass these restrictions, allowing them to communicate freely, access unbiased information, and maintain privacy in environments where digital rights are under threat[4].

These tools are vital for activists, journalists, and marginalized peoples operating under oppressive regimes, where censorship is often wielded to suppress dissent and hinder social movements[4]. By enabling secure and anonymous communication, evasion tools play a crucial role in mobilizing protests, sharing information,

and holding authorities accountable[1]. However, these technologies can be misused for illicit activities, such as illegal trade or harmful cyber activities, complicating their regulation and legal status[2]. The dual nature of censorship evasion tools presents a challenge: while they are essential for supporting human rights and freedom of expression, they also require consideration of the broader societal impacts and potential for misuse they present. This makes their development and deployment a complex issue[5].

III. MOTIVATION

The motivation for this project is driven by the growing trend of internet censorship, particularly in authoritarian countries. The Internet, once seen as a free and open platform for information exchange and communication, is increasingly being controlled and restricted by governments aiming to suppress dissent and control the flow of information. These restrictions threaten freedom of speech, access to unbiased information, and personal privacy.

In response to these censorship tactics, various tools, including VPNs, Tor, and encrypted communication methods, have been developed to help users bypass restrictions and maintain open access to the Internet. However, as these tools evolve, so do government countermeasures, leading to a complex battle between those seeking to preserve internet freedom and those imposing censorship.

This project seeks to analyze the technical and socio-political dimensions of censorship evasion tools, assessing their effectiveness, limitations, and ethical considerations. By examining these tools within high-censorship regions such as China, Iran, and Russia, the study aims to provide a comprehensive understanding of how these tools operate, how they are countered by governments, and what broader impacts they have on national security, privacy, and free expression.

IV. RESEARCH SIGNIFICANCE

This research is critical for several reasons. First, it provides a technical analysis of how censorship evasion tools function and evaluates their effectiveness against the countermeasures used by authoritarian regimes. This

insight is necessary for improving these tools' resilience and adaptability in restrictive regions. Second, the study has broad geopolitical relevance, offering insights into how censorship affects societies under restrictive regimes and how these tools support fundamental human rights like free expression. Additionally, the project addresses complex ethical and legal questions, considering the dual-use nature of these tools, which can be misused for illegal activities. The study's findings will support policy recommendations that balance national security concerns with the need to uphold digital rights. This will contribute to a global understanding of how to protect information access in an increasingly controlled internet landscape.

Existing Literature and Research Gaps: Summarize key findings from existing research on censorship evasion and the limitations of current tools. Identify gaps in the literature, such as limited data on the real-world effectiveness of specific tools in different countries, which your study aims to address.

V. METHODOLOGY

This study employs a methodology consisting of a literature review, tool evaluation, case studies, technical analysis, and an ethical and legal assessment. The objective is to thoroughly investigate the capabilities, limitations, and uses of censorship evasion tools across different high-censorship regions. Below is a detailed outline of each methodological step.

A. Technical Analysis

An in-depth technical analysis is conducted to understand the underlying mechanisms of censorship evasion tools. This involves examining encryption protocols, proxy networks, and obfuscation techniques using network analysis tools such as Wireshark. The traffic signatures of each tool are analyzed to assess how effectively they blend into regular internet traffic and evade detection by DPI systems.

B. Tool Evaluation and Comparative Analysis

This study conducts an evaluation and comparative analysis of the most widely used censorship evasion

tools, such as VPNs, Tor, and Psiphon. These tools are assessed based on the following criteria:

- **Effectiveness:** Each tool's ability to bypass common censorship mechanisms, such as DNS filtering and geoblocking, is tested in controlled environments. Metrics include success rates in accessing blocked content, connection stability, and speed.
- **Performance:** Using tools like Speedtest.net, we measure the impact of each censorship evasion tool on internet speed and latency, evaluating usability under various conditions, including streaming and large file downloads.
- **Leak and Privacy Tests:** DNS, IP, and WebRTC leak tests are performed using platforms such as DNSLeakTest.com and BrowserLeaks.com to evaluate each tool's ability to maintain user anonymity and privacy.
- **Obfuscation Capabilities:** The obfuscation techniques of each tool are examined, including obfs4 for Tor and stealth servers for VPNs, to assess their ability to bypass DPI in restrictive environments.

VI. CASE STUDIES OF CENSORSHIP EVASION IN HIGH-RESTRICTION COUNTRIES

To put the technical findings in context, this section presents case studies examining censorship and the usage of censorship evasion tools in three countries: China, Iran, and Russia. By analyzing these examples, we can gain insight into the specific methods of censorship employed, the effectiveness of various tools in circumventing these restrictions, and the broader consequences of these tools. Each case study includes three primary components: censorship methods, tool effectiveness, and socio-political impact.

A. China

China is often considered one of the most restrictive environments for internet freedom. The country's comprehensive censorship system, known as the Great Firewall, combines filtering, content blocking, and sophisticated traffic analysis to control the flow of information.

- **Censorship Methods:** The Great Firewall employs DNS tampering, IP blocking, keyword filtering, and DPI to monitor and censor internet traffic. These

methods are enforced not only through technology but also by legal and corporate pressures on domestic platforms to comply with government regulations[6]. Social media and messaging apps are routinely monitored, and sensitive topics, such as political dissent or information about human rights, are heavily censored.

- **Tool Effectiveness:** While the Chinese government continuously adapts to block popular evasion tools, VPNs, proxy services, and the Tor network remain among the most widely used methods for bypassing censorship. The effectiveness of these tools varies; VPNs, for instance, are often targeted and disabled by government countermeasures, while Tor traffic is sometimes detected and blocked by DPI systems. However, tools that use obfuscation techniques, such as Shadowsocks, have seen success by disguising traffic as regular HTTPS traffic, making it harder for the Firewall to detect[6].
- **Socio-Political Impact:** Censorship evasion in China is critical in preserving some degree of online freedom for citizens and activists. These tools enable individuals to access banned information, participate in discussions on sensitive topics, and coordinate activism. For instance, during events like the 2019 Hong Kong protests, censorship evasion tools facilitated communication and allowed citizens to share real-time updates[6]. However, government pressure to limit VPN access and the use of legal consequences for bypassing censorship create significant risks for users.

B. Iran

Iran faces intermittent and severe internet restrictions, particularly during political unrest. Internet censorship in Iran combines government-mandated restrictions with strategic internet shutdowns to limit access to information and suppress protests.

- **Censorship Methods:** Iran's approach to censorship is multifaceted, including URL filtering, social media blocks, and national firewalls. The government frequently disrupts internet service entirely during protests or politically sensitive periods, implementing near-total blackouts in some areas. The

authorities also mandate that domestic apps and services follow strict censorship guidelines, limiting users' access to non-government-approved content[7].

- **Tool Effectiveness:** Given the unpredictable nature of restrictions, tools such as VPNs and Tor play a significant role in enabling internet access in Iran. However, these tools face frequent disruptions, with VPN providers often targeted by government crackdowns. Some users have turned to tools that leverage peer-to-peer connections or offline sharing to circumvent complete blackouts, though these solutions are less reliable and more technically complex. During extensive shutdowns, satellite internet and mesh networks have been explored as alternatives, although they are not widely accessible[8].
- **Socio-Political Impact:** Censorship evasion tools are crucial for maintaining lines of communication during protests, especially when internet shutdowns are in place. They allow Iranians to document and disseminate evidence of government actions and facilitate international awareness. The socio-political impact of censorship evasion is thus significant, fostering digital activism and providing channels for citizen journalism[9]. However, these tools can also place users at risk of detection and punishment, given the Iranian government's extensive surveillance apparatus.

C. Russia

In recent years, Russia has intensified its control over the Internet, implementing laws and technical measures that limit access to information and increase surveillance of Internet users.

- **Censorship Methods:** Russian censorship practices include IP and URL blocking, government-mandated data storage for social media platforms, and DPI to monitor and restrict internet traffic. The country's "sovereign internet" project aims to create a national internet infrastructure that could operate independently from the global Internet, thereby providing the government with granular control over domestic traffic[10]. Popular social media platforms

and messaging services that refuse to cooperate with data regulations are often throttled or blocked.

- **Tool Effectiveness:** To navigate these restrictions, Russians increasingly rely on VPNs, Tor, and proxy servers. However, new laws criminalizing the distribution and use of unapproved VPNs have made access to these tools more challenging. Telegram, a popular messaging app with built-in evasion mechanisms, remains widely used despite government attempts to block it due to its resilience against censorship. Nevertheless, as government regulations and technical restrictions evolve, the long-term efficacy of these tools remains uncertain.
- **Socio-Political Impact:** In Russia, censorship evasion tools support political dissent by allowing activists and opposition groups to communicate and mobilize. These tools provide access to information from independent media sources otherwise inaccessible within the country. Censorship evasion has become particularly relevant after recent protests, as citizens seek to circumvent media restrictions and document government responses. However, the increasing pressure on these tools suggests a narrowing space for online activism and a heightened risk for users involved in socio-political movements [11].

These case studies demonstrate the complexities of censorship and the dynamic nature of evasion strategies in high-restriction environments. While censorship evasion tools offer critical access to information and support for activism, they face substantial obstacles and risks that vary according to each country's unique socio-political landscape [12].

VII. ETHICAL AND LEGAL IMPLICATIONS OF CENSORSHIP EVASION TOOLS

While essential for promoting internet freedom in restrictive environments, the development and use of censorship evasion tools also raise ethical and legal questions. This section explores the responsibilities of developers, policymakers, and users in navigating the dual purpose of these tools: enabling access to information and protecting privacy on the one hand and the potential for misuse in illegal activities on the other. We

review existing legal frameworks and policies in high-censorship regions, alongside ethical guidelines from internet freedom advocacy organizations, to examine the balance between facilitating open access and addressing national security concerns.

A. Ethical Considerations

The ethical considerations surrounding censorship evasion tools are complex, as these tools empower citizens to exercise their right to information and communication, particularly in environments with limited freedoms. For activists, journalists, and ordinary users, these tools provide critical access to unbiased information, secure communication channels, and a platform for socio-political engagement. However, using these tools is not without risk; many users may face severe repercussions if detected, including legal penalties, detention, or worse[13].

From an ethical perspective, developers of censorship evasion tools are responsible for minimizing user risks by enhancing the security, reliability, and accessibility of these tools. These developers must also consider potential misuse, as evasion tools can be used for activities like illicit trade, hacking, or other illegal purposes. Balancing this ethical tension involves implementing responsible design practices, such as educating users on secure usage and building features to help limit potential harm without compromising core freedoms[14].

Internet freedom advocacy organizations, such as Access Now and the Electronic Frontier Foundation, advocate for responsible use guidelines, emphasizing the importance of transparency, user consent, and security-first development practices. These organizations also encourage developers to work collaboratively with local communities to ensure that tools are culturally and technically suited to their deployed contexts.

B. Legal Frameworks and Policy Implications

The legal status of censorship evasion tools varies widely depending on the country. In high-censorship regions such as China, Iran, and Russia, legal frameworks typically prohibit or severely restrict such tools, viewing them as threats to national security and social stability. For instance, China criminalizes the unauthorized use

of VPNs, while Iran requires VPN providers to be government-approved, effectively restricting access to non-compliant services. Russia has implemented laws that ban unregistered VPNs and even criminalize the promotion of these tools, which complicates user access and developers' operations[15].

The policies in these countries reflect the tension between security and access. Policymakers argue that censorship is necessary to combat misinformation, terrorism, and other security threats; however, these restrictions often also target politically sensitive content and independent journalism, blurring the line between security and suppression. This legal landscape places developers, providers, and users in challenging positions, as accessing or distributing censorship evasion tools can be met with serious legal consequences.

Policymakers in less restrictive countries face different ethical and legal challenges. While they may not impose bans, they are often called upon to consider the implications of supporting or restricting access to evasion tools internationally, especially when national security or foreign policy concerns intersect with digital rights advocacy. Some countries have enacted policies that encourage the development of censorship evasion tools to promote internet freedom abroad. Still, these efforts must also consider diplomatic ramifications and ensure compliance with international laws.

C. Responsibilities of Developers and Policymakers

Developers and policymakers each play critical roles in shaping the ethical and legal landscape of censorship evasion tools. Developers are responsible for designing tools prioritizing security, transparency, and usability while considering the legal and ethical implications in high-censorship environments. They can mitigate potential misuse by incorporating educational resources, security guides, and technical safeguards, supporting responsible use without limiting core functionalities. Collaboration with advocacy organizations and local users can help ensure these tools remain practical and ethical under challenging conditions.

On the other hand, policymakers are tasked with creating legal frameworks that balance the right to information access with national security needs. In high-

censorship regions, restrictive policies can make it difficult for developers to operate openly and for users to access essential tools safely. However, governments in freer countries can contribute to a global environment that respects security and freedom by enacting policies promoting transparency, security standards, and ethical guidelines. This approach also involves supporting internet freedom initiatives and encouraging cross-border cooperation to advance ethical standards in censorship evasion.

D. Guidelines from Internet Freedom Advocacy Organizations

Internet freedom advocacy organizations provide valuable ethical guidelines and best practices for developers, policymakers, and users. Organizations like the Open Technology Fund, Access Now, and the Tor Project emphasize the importance of security, privacy, and transparency in developing and deploying censorship evasion tools. Their guidelines advocate for end-to-end encryption, user education, and community-informed development processes to ensure these tools effectively meet users' needs in high-censorship environments while mitigating potential risks.

These organizations also support policy frameworks prioritizing user rights, encouraging transparency, and reducing misuse risks. For example, they may recommend policies that protect developers from prosecution for creating censorship evasion tools or advocate for international protections for users who face legal consequences for accessing information. By fostering an ethical and legally supportive environment, these organizations contribute to advancing internet freedom.

In summary, the ethical and legal implications of censorship evasion tools require careful consideration by developers, policymakers, and advocates alike. A balanced approach that promotes secure and accessible tools, transparent policies, and ethical development practices is essential to support internet freedom while addressing security and legal concerns.

VIII. RESULTS

By manually testing the internet speed test by using tools such as speedtest.net for bandwidth metric

and privacy testing through using ipleak.net, browserleaks.net, and dnsleak.net, we were able to obtain crucial information and data to evaluate the functionality and performance of NordVPN and Psiphon. These tests were conducted under different conditions, such as the different connection locations under three different connection protocols. We will show the collected data and our static analysis of these data. The output data will provide crucial insight on the connection performance.

A. Performance analysis on NordVPN

To have a successful analysis of the performance of NordVPN, we used speedtest.net to collect real-time data on essential elements involving the network's speed and quality, such as Down/Up speeds with Ping and Up/Down latency. We conducted three tests: VPN to Dallas, Hong Kong, and Nord's Obfuscation server. In addition, for each test, we record all the data involving different types of connection, such as OpenVPN-UDP, OpenVPN-TCP, and NordLynx. OpenVPN-UDP stands for User Datagram Protocol where it does not rely on an established dedicated server to send network packet. OpenVPN-TCP stands for Transmission Control Protocols where there is an established dedicated connection between the communicators. In comparison, UDP will prioritize speed over reliability which is the complete opposite [16]. NordLynx is a new generation VPN protocol developed by NordVPN inspired by WireGuard's VPN protocol.

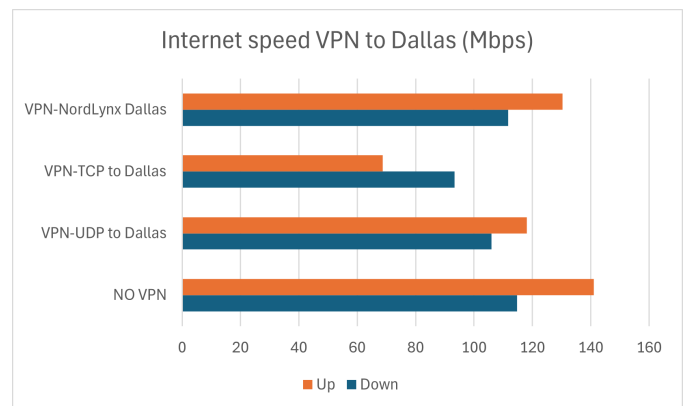


Fig. 1: Internet speed test Nord to Dallas(Mbps)

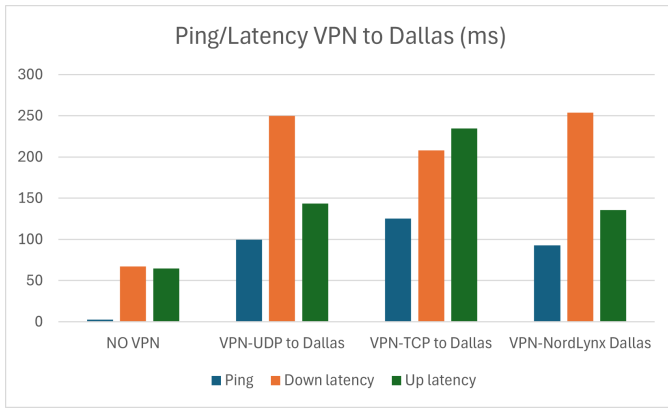


Fig. 2: Ping/Latency Nord to Dallas(ms)

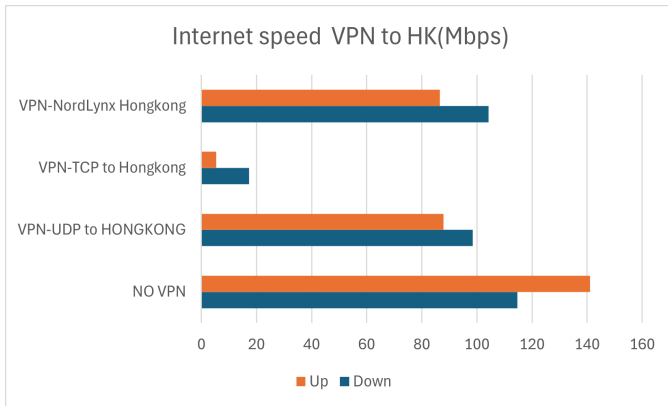


Fig. 3: Internet speed Nord to HK(Mbps)

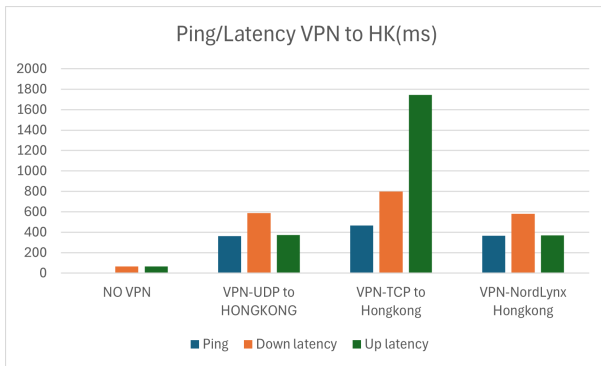


Fig. 4: Ping/Latency Nord to HK(ms)

Figure 1, 2, 3, 4, each demonstrates the internet performance when the system is not connected to a VPN to when it does with three different types of protocols to Dallas and Hong Kong respectively. The reason behind the Dallas and Hong Kong locations is that Dallas offers insight into the fluctuation in a closer range. Hong Kong provides information on a further area surrounded by countries with high-level censorship restrictions.

For Dallas, across three different types of connection protocols, there is little to no change in the internet speed for both Up and Down except for the TCP protocol. In the Download speed, there is only 7.5% decrease for UDP, 18.6% for TCP, 2.7% for NordLynx. For the Up speed, 16.2% UDP, 51.24% TCP, and 7.6% NordLynx. Additionally, despite the little change in internet performance with UDP and NordLynx protocols compared to TCP, latency performance is significantly decreased. There is an increase of 25 to 44 times in the duration of packets traveling back and forth from devices to the destination server. This will affect users' ability to perform daily tasks such as web browsing, video streaming, gaming, etc. For Hong Kong, the internet performance change is significantly worse compared to the connection to Dallas. For UDP and NordLynx, there is a 14% 9% decrease in the down speed and 38%

For both connections across three different types of VPN protocols, NordVPN did a great job in protecting users' IP, DNS, and WebRTC from getting leaked. Zero forwarded IPs were detected throughout the entire testing process. However, for the DNS leak test from Dallas and Hong Kong, we detected five servers, each from NordVPN ISP and PacketHub ISP. Despite these DNS records being "leaked," it does not threaten your privacy. NordVPN is known to have 6700+ encrypted servers to protect users from leaking their DNS; they use their servers to send DNS queries in an encrypted approach so that users' DNS won't be exposed [17]. PacketHub and NordVPN are known to be a part of NordVPN encrypted server. These VPN connections show that a third-party application is interfering with the connection, as it is easy to recognize when a VPN is being used. This is when an Obfuscation method comes into play.

Figure 5 and figure 6 demonstrate the performance of NordVPN's obfuscation techniques. NordVPN's obfuscation technique allows users to go undetected by outside sources if they use a VPN. However, with the obfuscation involvement, it only provides a connection to the best-performance server based on the user's location. If they want to evade a certain censorship restriction, it is unlikely to achieve so. Since it provides the best possible server for connection, the obfuscation performance was

only impacted slightly. For download speed, less than 10% decrease in speed across three different protocols.

On the latency side, the effect on the performances was consistent across all three protocols. In the ping aspect, the performance experienced a degradation by over 30 times compared to the original; however, up and down latency only got affected by around 50%

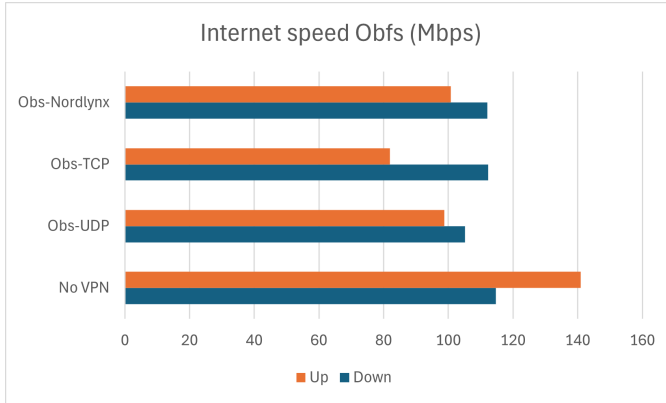


Fig. 5: Internet speed Obfs (Mbps)

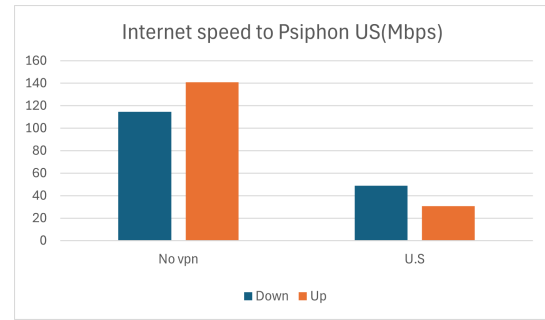


Fig. 7: Internet speed to Psiphon US(Mbps)

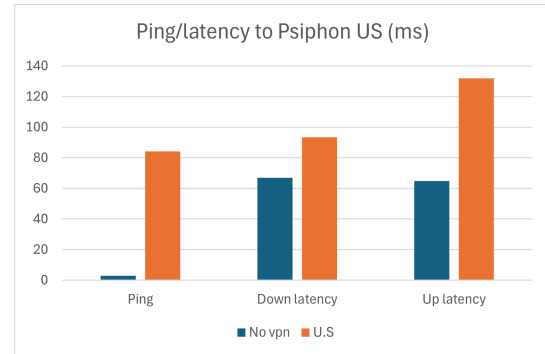


Fig. 8: Ping/latency to US (ms)

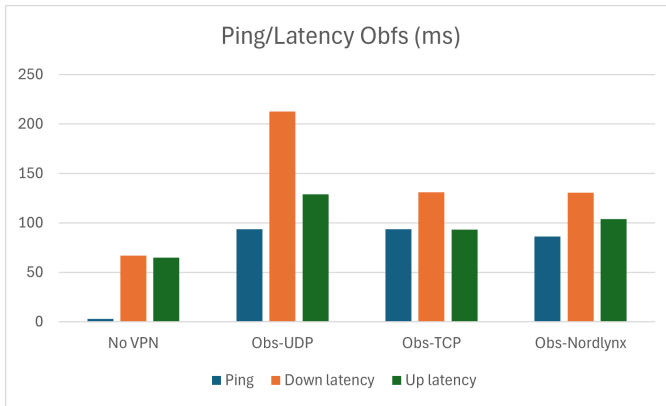


Fig. 6: Ping/Latency Obfs (ms)

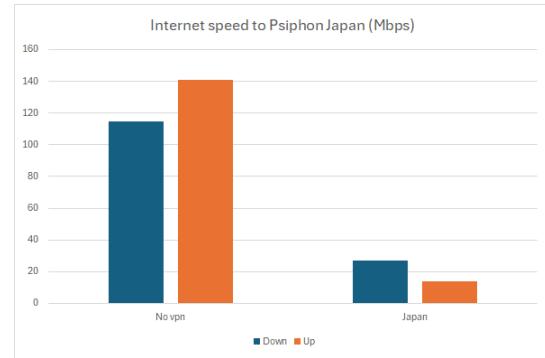


Fig. 9: Internet speed to Psiphon Japan (Mbps)

B. Performance analysis on Psiphon

Initially, we wanted to match our testing location with our NordVPN (Dallas and Hong Kong); however, due to Psiphon limitation in the location option, we were only able to perform two different connection tests, one to the US and one to Japan. Up and down speed suffers significantly as the download speed decreases by roughly 60%

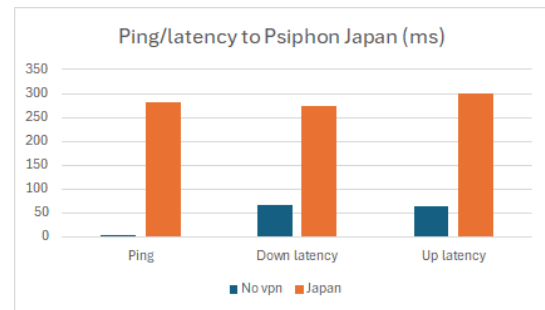


Fig. 10: Ping/latency to Japan (ms)

C. Performance analysis on Tor

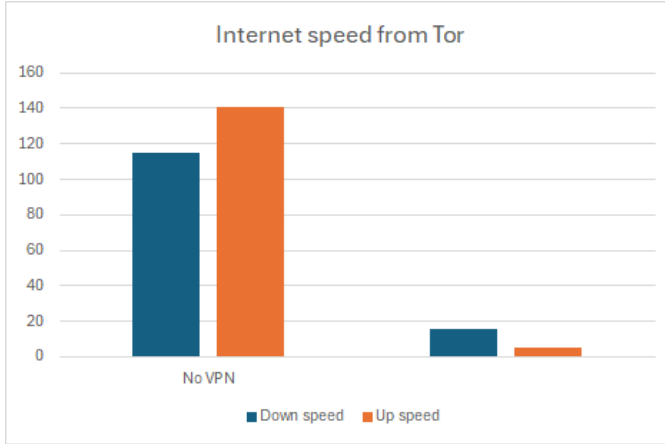


Fig. 11: Tor internet speed

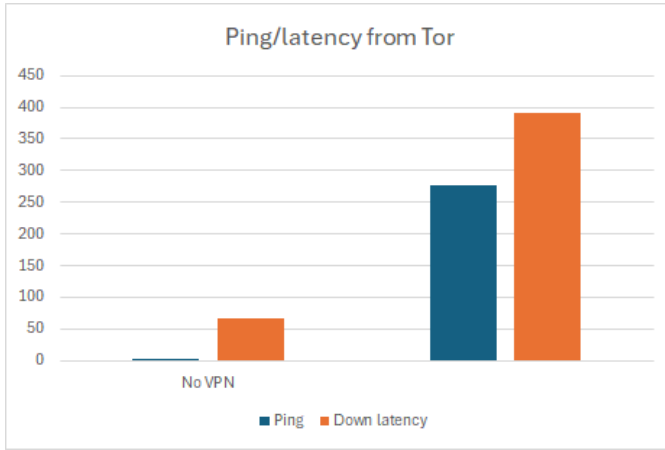


Fig. 12: Tor latency

The speed changes while using Tor compared to the standard internet connection were significant. The download speed experienced a decrease of over 86%. Meanwhile, the upload speed saw a substantial dropoff with 96%. The latency changes were also drastic as the ping increased by a factor of 99, and the download latency increased by a factor of 6. Unfortunately, we could not consistently record the upload latency because the number exceeded the threshold. In terms of privacy, there are no IP and DNS leaks. There are no WebRTC leaks since the Tor browser does not have built-in WebRTC capabilities. In addition, Tor provides an entirely different IP address for each connection established, enhancing the user's anonymity.

D. NordVPN vs Psiphon

Table II shows the differences between NordVPN and Psiphon.

Even though externally, Psiphon as an application serves the same purpose as NordVPN, it was created with different priorities. NordVPN was created as a privacy/security tool, and censorship evasion is just an add-on feature. Psiphon is an open-source internet censorship circumvention product made primarily to evade censorship in heavily restricted environments. Unfortunately, their focus on censorship evasion negatively affected the connection performance and security aspects.

When using Psiphon, a WebRTC leak test may reveal that the WebRTC IP differs from the assigned remote IP of Psiphon, alerting a potential leak. WebRTC stands for Web Real-Time Communication, allowing browsers and mobile applications to have audio/video calling, chat, and P2P file sharing without being dependent on plugins or third-party software [18]. If a VPN service fails to handle network traffic properly, especially WebRTC traffic, users' IP addresses can be exposed. This issue occurs in our WebRTC testing. Although Psiphon hides the local IP, it may not mask WebRTC IP requests, potentially exposing information such as the ISP-assigned IP and general location.

During the use of Psiphon, multiple DNS services may be detected because Psiphon is known to use a variety of proxy servers from VPN to SSH to HTTP/HTTPS, all depending on the network conditions and the restriction level of censorship [19]. This can result in multiple DNS requests, considering Psiphon's dynamic servers routing capability, or use split tunneling for specific traffic. None of the DNS detected contains our IP.

In contrast, NordVPN has its own private encrypted DNS server. When a connection is established, NordVPN directs all DNS requests through its own DNS infrastructure, which provides a safe and consistent connection toward the DNS requests. It reduces the chance of DNS leaks. This setup prioritizes privacy and security by keeping all traffic, including DNS requests, within the secure VPN tunnel [17].

E. Wireshark interception

File Name	Total	TCP	UDP
VPN_HK_lynx1.pcapng	4018	0	4016
VPN_HK_lynx2.pcapng	3919	0	3917
VPN_HK_lynx3.pcapng	3950	0	3948
VPN_HK_lynx4.pcapng	3939	0	3934
VPN_HK_tcp1.pcapng	3894	3873	19
VPN_HK_tcp2.pcapng	3882	3846	35
VPN_HK_tcp3.pcapng	4355	4334	21
VPN_HK_tcp4.pcapng	4570	4550	20
VPN_HK_UDP.pcapng	4040	1	4037
VPN_HK_UDP2.pcapng	3968	0	3967
VPN_HK_UDP3.pcapng	4010	0	4005
VPN_HK_UDP4.pcapng	3970	1	3969
VPN_lynx.pcapng	4031	9	4020
VPN_lynx2.pcapng	1206	0	1203
VPN_lynx3.pcapng	3939	0	3939
VPN_lynx4.pcapng	3990	0	3988
VPN_tcp1.pcapng	4578	4561	16
VPN_tcp2.pcapng	3771	3746	24
VPN_tcp3.pcapng	3837	3819	18
VPN_tcp4.pcapng	3862	3832	30
VPN_UDP.pcapng	4549	15	4531
VPN_UDP2.pcapng	4061	0	4059
VPN_UDP3.pcapng	3827	0	3825
VPN_UDP4.pcapng	4279	0	4277
VPN_Obf_lynx1.pcapng	4024	0	4023
VPN_Obf_lynx2.pcapng	4200	0	4199
VPN_Obf_lynx3.pcapng	4181	0	4178
VPN_Obf_lynx4.pcapng	4127	0	4125
VPN_Obf_tcp1.pcapng	4254	4236	17
VPN_Obf_tcp2.pcapng	3820	3783	36
VPN_Obf_tcp3.pcapng	4147	4123	23
VPN_Obf_tcp4.pcapng	3914	3888	26
VPN_Obf_udp1.pcapng	3975	0	3971
VPN_Obf_udp2.pcapng	3859	0	3857
VPN_Obf_udp3.pcapng	3991	0	3991
VPN_Obf_udp4.pcapng	3943	0	3943

TABLE I: Packet Analysis Data

Before our testing, we devised a hypothesis on what we expect with our future findings. Our hypotheses are:

- With each connection from a certain protocol, the majority of packets captured will have that protocol. Ex: OPENVPN-UDP has the majority of UDP packets, OPENVPN-TCP has the majority of TCP packets, NordLynx has the majority of UDP/Wire-guard packets
- Similar trends are expected across different location connections.

First, we manually captured the network traffic across different protocols for at least 10 seconds. We did it 4 times for each protocol. Then, we put all the ".pcapng" files in a folder. created a python script PacketCaptureAnalyzer.py that are capable of traversing through all the ".pcapng" files inside a folder and use the two tshark commands "tshark -r <file name>.pcapng | wc -l" and "tshark -r <file name>.pcapng -Y "<protocol type>" -T fields -e frame.number | wc -l" to capture the total packets, the total UDP packets, and the total TCP packets.

Table I shows the results after we run the script. Each row displays the processed file, representing the breakdown of packet counts for total, TCP, and UDP packets. This result proves our hypothesis is correct. For both UDP and Nordlynx protocols, despite the different locations, the number of packets captured is dominated by UDP protocols. More than 99% of the captured packets are using the UPD protocol. A similar trend can be observed in the type of TCP connection. Across different locations, over 99% of packets captured are under TCP protocols.

User Expectations Users of censorship evasion tools often form expectations based on these technologies' marketing claims and reputations. Each tool under evaluation—Tor, NordVPN, and Psiphon—projects certain key features that appeal to users in restrictive environments. However, discrepancies between advertised features and real-world performance often shape user perceptions.

Tor is widely recognized for its robust anonymity and decentralized routing capabilities. Its marketing emphasizes these strengths, which resonate well with users seeking to preserve privacy and evade surveillance. However, users often overestimate its usability for real-time applications like video calls or gaming. Tor's inherently slow speeds and high latency, driven by its multi-node routing and encryption processes, are less emphasized in its promotion, leaving users unprepared for these limitations.

NordVPN positions itself as a high-security and high-performance solution. Its marketing highlights advanced

Feature	NordVPN	Psiphon
1. Bypassing Censorship	- A high performance of protocol options with strong privacy and security capability. The obfuscation technique performs well as it masks VPN traces. - Stable connections.	- More flexibility in the provided protocols (SSH, HTTP proxy, etc..) but lack of reliability. - Inconsistent connection.
2. Speed and Performance	- Provide more up-to-date protocols such as Nord-Lynx that ensure faster and more stable performance.	- Outdated protocols and system leads to slower speeds and higher latency.
3. Privacy and Security	- Strong OpenVPN AES-256 encryption with zero log recording. [3]	- Simple encryption; known to collect data for analytics purposes. [20]
4. Cost	- Paid subscription with high-value features and premium performance.	- Free version available; premium version is cheaper but offers limited performance.

TABLE II: Comparison of NordVPN and Psiphon

features such as strong encryption, obfuscation, and fast speeds. These claims align with the tool’s capabilities, but NordVPN’s marketing tends to oversimplify its challenges in high-censorship regions. For example, while NordVPN’s obfuscation servers work effectively in many cases, their performance can vary depending on the intensity of censorship measures. Additionally, occasional DNS leaks—though not privacy-compromising—are rarely addressed in promotional materials, leading users to expect flawless security under all conditions.

On the other hand, Psiphon emphasizes its accessibility and censorship circumvention capabilities, particularly its ability to adapt to restrictive environments through diverse protocols like VPN, SSH, and HTTP proxy. While this focus is accurate, Psiphon often downplays its shortcomings in privacy and performance. Unlike NordVPN or Tor, Psiphon is less effective at protecting user anonymity and is prone to performance issues such as reduced speeds and increased latency. Users expecting seamless and secure browsing may find Psiphon unsuitable for high-risk activities requiring strong encryption and privacy.

A key gap across all tools is the level of transparency regarding their limitations. While each tool provides accurate summaries of its features, they tend to underrepresent areas like speed reductions, privacy trade-offs, or reliability in extreme censorship scenarios. This creates a potential mismatch between user expectations and actual experiences.

IX. CONCLUSION

This study highlights the capabilities and limitations of censorship evasion tools, focusing on NordVPN, Tor, and Psiphon and their performance in high-censorship regions such as China, Iran, and Russia. While these tools effectively promote internet freedom and privacy, they face significant challenges, including performance degradation, evolving government countermeasures, and transparency gaps in user expectations.

The ongoing battle between censorship and evasion underscores the need for adaptive, resilient technologies and collaborative efforts among developers, policymakers, and advocacy groups. Context-specific designs and robust security enhancements are critical to addressing users’ diverse challenges in restrictive environments.

Censorship evasion tools remain vital for preserving access to information and supporting activism in authoritarian regimes. However, balancing these tools’ benefits with ethical and legal considerations is essential. This study provides a foundation for further research into improving these tools and addressing internet censorship’s complex socio-political and technological challenges.

X. FUTURE WORK

The rapidly evolving landscape of internet censorship and evasion techniques presents numerous avenues for future research. This section outlines potential directions to build upon this study’s findings and address the limitations and challenges identified.

A. Development of Adaptive and Resilient Evasion Tools

One promising direction for future research is the development of more adaptive censorship evasion tools. Current tools, such as VPNs and Tor, face constant countermeasures from government authorities. Future work could explore using artificial intelligence and machine learning to create evasion tools that dynamically adapt to real-time network conditions and evolving censorship tactics, including DPI and AI-driven traffic analysis. By integrating these adaptive mechanisms, future tools may be able to respond to censorship patterns and obfuscate traffic in novel ways, increasing resilience in highly restrictive environments.

B. User-Centric Design and Accessibility Improvements

The usability of censorship evasion tools remains critical, particularly for non-technical users in high-restriction regions. Future research could investigate user-centered design approaches to improve the accessibility and intuitiveness of these tools, particularly in regions with low digital literacy. This includes exploring lightweight versions of tools optimized for lower bandwidth environments, local language support, and user education initiatives to enhance understanding of secure usage practices. Improving user experience can make these tools more widely adopted and less likely to be misconfigured, enhancing their effectiveness and security.

C. Longitudinal Studies on Socio-Political Impacts

While this study provides a snapshot of the socio-political impact of censorship evasion tools, a longer-term analysis could yield more profound insights into how these tools influence political and social dynamics over time. Future research could focus on longitudinal studies that track the effects of censorship evasion on activism, information dissemination, and public opinion in restrictive regimes. Such studies could involve monitoring changes in internet censorship policies and evaluating how sustained access to evasion tools shapes public awareness, resistance to censorship, and overall political climate.

D. Legal and Ethical Frameworks for Dual-Use Technologies

The dual-use nature of censorship evasion tools—facilitating free access to information while being potentially misused for illegal activities—presents complex ethical and legal challenges. Future work could explore the development of legal and ethical frameworks tailored to dual-use technologies, balancing support for internet freedom with concerns around national security. This could involve collaboration between international legal bodies, human rights organizations, and technology developers to establish guidelines that protect developers and users while addressing the risks associated with misuse.

E. Integration with Emerging Technologies

Future research could investigate integrating censorship evasion tools with emerging technologies like blockchain and decentralized web infrastructures. For instance, blockchain-based DNS and content delivery networks could offer alternative pathways to access information beyond traditional internet structures, potentially circumventing censorship. Additionally, further exploration of peer-to-peer networks and mesh networking solutions may provide viable options for bypassing internet shutdowns and localized blockages, offering users more decentralized and resilient communication.

F. Enhanced Security and Privacy in High-Risk Environments

Given the risks users face when evading censorship in authoritarian regions, enhancing the security and privacy of censorship evasion tools remains a priority. Future work could delve into advanced encryption protocols, multi-hop routing, and zero-knowledge architectures that minimize metadata exposure and make it harder for authorities to identify users. Such research could also explore methods for obfuscating traffic more effectively to avoid detection by sophisticated surveillance systems, offering increased protection for high-risk users.

G. Cross-Border Collaboration and Knowledge Sharing

The global nature of internet censorship calls for cross-border collaboration among developers, activists,

policymakers, and researchers. Future research initiatives could focus on building networks for knowledge sharing and resource pooling, creating platforms where stakeholders can exchange insights, tools, and best practices. This could foster a collaborative environment for refining censorship evasion strategies, developing more robust tools, and advocating for global standards in digital rights protection.

By pursuing these research directions, the field can continue to evolve in response to the complex challenges of censorship and internet freedom, ultimately advancing the effectiveness, accessibility, and ethical deployment of censorship evasion tools.

XI. TEAM BREAKDOWN

- J. Vanlyssel
 - Literature Review and Background Research
 - Case Studies on High-Restriction Countries
 - Ethical and Legal Assessment
 - Results Compilation and Analysis
 - Writing and Document Preparation
- L. Su
 - Literature Review and Background Research
 - Technical Analysis and Tool Testing
 - Results Compilation and Analysis
 - Writing and Document Preparation

XII. REFERENCES

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